

JDBC Setup and Connection to spotify_clone

Q1: Download Connector/J and add it to your IDE classpath

Download

1. Go to dev.mysql.com/downloads/connector/j/.
2. Under "Select Operating System", choose **Platform Independent**.
3. Download the ZIP, extract it, and keep the file `mysql-connector-j-x.x.x.jar` in a folder such as `C:\jdbc`.

Eclipse

1. Right-click your project and choose **Build Path > Configure Build Path**.
2. Open the **Libraries** tab, select **Classpath**, and click **Add External JARs...**
3. Select `mysql-connector-j-x.x.x.jar`, then click **Apply and Close**.
4. The jar now appears under **Referenced Libraries**.

IntelliJ IDEA

1. Go to **File > Project Structure > Modules (or Libraries)**.
2. Click **+** and choose **Java**, then select the jar file.
3. Click **OK** and then **Apply**.

NetBeans

1. Right-click **Libraries** in your project and choose **Add JAR/Folder...**
2. Select the jar file and click **Open**.

Check: the jar should now be listed under Referenced Libraries (Eclipse), External Libraries (IntelliJ) or Libraries (NetBeans).

Create the database first in MySQL:

```
CREATE DATABASE spotify_clone;
```

Q2: SpotifyDBTest.java (basic connection)

```
// SpotifyDBTest.java
// Connects to the MySQL database 'spotify_clone' using DriverManager.getConnection().

import java.sql.Connection;
import java.sql.DriverManager;

public class SpotifyDBTest {
    public static void main(String[] args) throws Exception {
        String url      = "jdbc:mysql://localhost:3306/spotify_clone";
        String username = "root";
        String password = "root123";

        Connection con = DriverManager.getConnection(url, username, password);

        if (con != null) {
            System.out.println("Connection Successful");
            con.close();
        }
    }
}
```

Output:

```
Connection Successful
```

Q3: Handling SQLException and ClassNotFoundException

```
// SpotifyDBTest.java
// Connects to 'spotify_clone' and handles errors with separate catch blocks.

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class SpotifyDBTest {
    public static void main(String[] args) {
        String url      = "jdbc:mysql://localhost:3306/spotify_clone";
        String username = "root";
        String password = "root123";

        try {
            // Load the MySQL JDBC driver class
            Class.forName("com.mysql.cj.jdbc.Driver");

            // Try to connect
            Connection con = DriverManager.getConnection(url, username, password);

            if (con != null) {
                System.out.println("Connection Successful");
                con.close();
            }
        } catch (ClassNotFoundException e) {
            // Occurs when the Connector/J jar is missing from the classpath
            System.out.println("ClassNotFoundException occurred: JDBC driver not found.");
            System.out.println("Details: " + e.getMessage());
        } catch (SQLException e) {
            // Occurs when the connection fails (wrong password, wrong URL, server down, etc.)
            System.out.println("SQLException occurred: Could not connect to the database.");
            System.out.println("Details: " + e.getMessage());
        }
    }
}
```

Output if the connection works:

```
Connection Successful
```

Output if the jar is missing from the classpath:

```
ClassNotFoundException occurred: JDBC driver not found.
Details: com.mysql.cj.jdbc.Driver
```

Output if the connection fails (for example, wrong password):

```
SQLException occurred: Could not connect to the database.
Details: Access denied for user 'root'@'localhost' (using password: YES)
```

Q4: Wrong password test

Change only the password line:

```
String password = "wrongpass";    // intentionally wrong
```

Exact error message you should see:

```
SQLException occurred: Could not connect to the database.
Details: Access denied for user 'root'@'localhost' (using password: YES)
```

The exact wording can differ slightly with your MySQL and driver version, so copy the message that appears on your own screen into your assignment.

One-line explanation: MySQL rejected the login because the password sent by the program did not match the password stored for the user root, so `getConnection()` threw a `SQLException` and no connection was made.